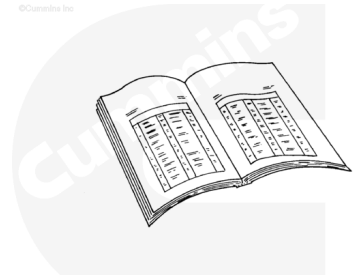


## Trainings ▾

### Preparatory Steps

#### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



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#### **⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**

#### **⚠ WARNING ⚠**

**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

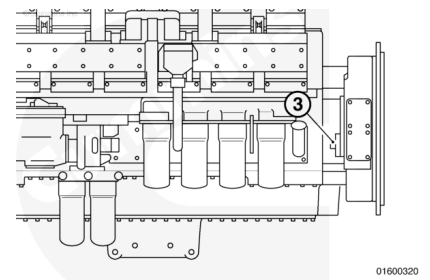
#### **⚠ WARNING ⚠**

**This component weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html>)
- Remove the full flow filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html>)

- Remove the lubricating oil pan adapter assembly. Refer to Procedure 007-017 in Section 7.  
(/qs3/pubsys2/xml/en/procedures/28/28-007-017.html)
- Remove the oil suction tube and brackets, if necessary, to access the connecting rod bearings. Refer to Procedure 007-040 in Section 7.  
(/qs3/pubsys2/xml/en/procedures/28/28-007-040-tr.html)

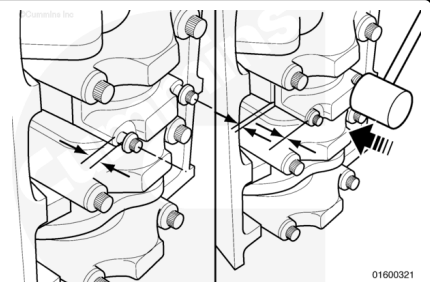
Use the barring mechanism (3) to rotate the crankshaft. Position the connecting rods near the bottom center for maximum accessibility when loosening the connecting rod capscrews.



## Remove

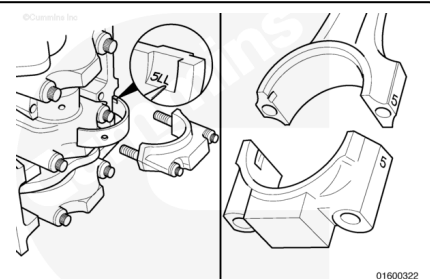
Loosen the capscrews until there is approximately 6.0 mm [0.24 in] of clearance between the cap and the capscrew head.

Use a soft-headed mallet to tap the capscrews alternately until the rod separates from the cap. Remove the capscrews and the cap.



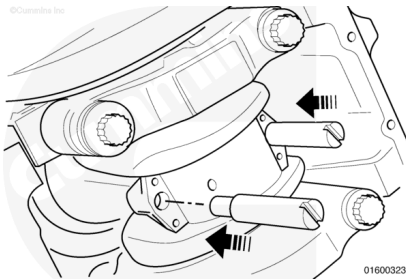
Remove the lower rod bearing. Scribe the bearing position in the tang area, as shown, for reuse or analysis.

**Note :** Connecting rods have their cylinder number and position stamped on the side of the rod and cap that is toward the camshaft. Check the rods and caps for correct markings.



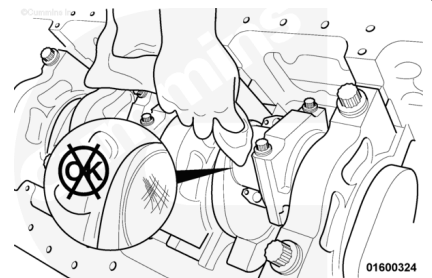
Push the rods up far enough to allow the upper bearings to be removed. Mark the cylinder number and location in the tang area, as described, for the lower bearing.

**Note :** Use of connecting rod guide pins, or equivalent, to prevent journal damage is recommended.



## Clean and Inspect for Reuse

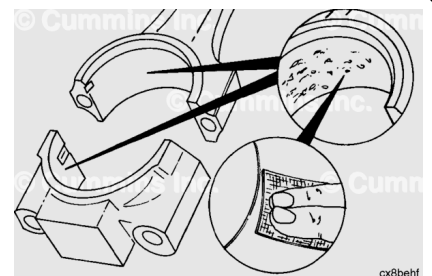
**Note :** If bearing replacement was initiated due to malfunction, check the oil pan for debris. If significant debris is present, remove and disassemble the lube oil pump and inspect the bushings for debris damage. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html>)



Clean the crankshaft journal with a lint-free cloth and inspect the journal for damage.

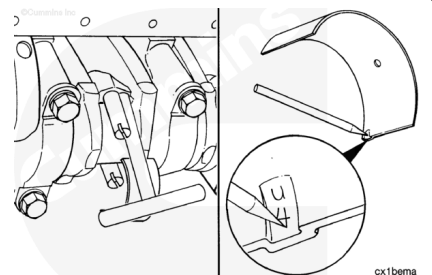
Clean the connecting rod and cap with a lint-free cloth. Inspect the rod cap, rod saddle, and capscrews for nicks, cracks, burrs, scratches, or fretting.

If nicks in the saddle area can **not** be removed with crocus cloth, the rod **must** be replaced.



The bearings **must** be marked for location as they are removed for future identification. Each bearing **must** be installed in its original location if the bearing is installed again.

Use a scribe or ball point pen to mark the bearing location in the tang area.

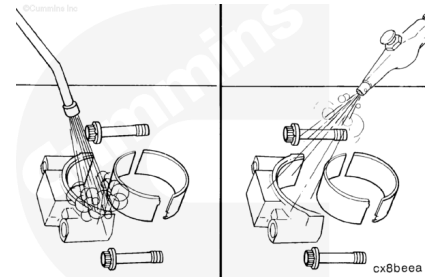


### **⚠ WARNING ⚠**

**When using solvents, acids or alkaline materials for cleaning, follow the manufacture's recommendations for use. Wear goggles and protective clothing to avoid personal injury.**

**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

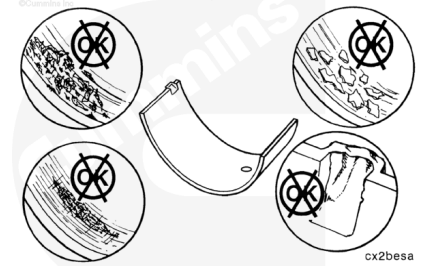


Use solvent to clean the bearings.

Dry with compressed air.

Inspect the bearing face on the front and back for:

- Nicks
- Cracks
- Burrs
- Scratches (deep enough to be felt with a fingernail)
- Scrapes
- Gouges
- Embedded particles
- Fretting
- Pitting
- Flaking
- Corrosion
- Lock tang damage.

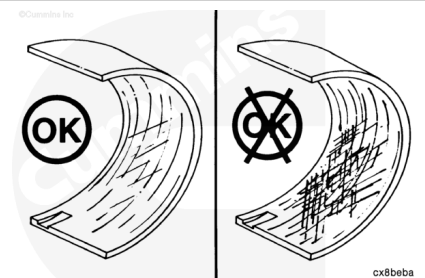


If any damage is found, the bearing **must** be replaced.

Normal bearing wear produces a smooth finish, which will wear into the copper lining. Exposed copper does **not always** indicate worn bearings. Reference the Parts Reuse Guidelines, Bulletin 3810303.

If large areas of copper lining are visible in the bearings before the engine has accumulated 3000 hours, inspect the engine for contamination from fine dirt particles and correct the problem.

For more detailed information of bearing damage, reference the Analysis and Prevention of Bearing Failures Manual, Bulletin 3810387.

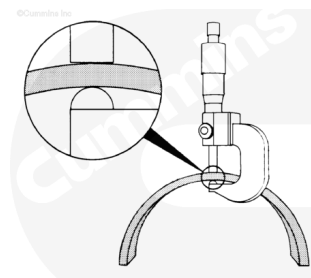


Use a ball-end micrometer to measure the thickness of the bearing in the wear location.

Replace the bearing when it is **not** within specifications.

## Connecting Rod Bearing Thickness Standard or Oversize

|                   | mm    |     | in     |
|-------------------|-------|-----|--------|
| Standard          | 3.106 | MIN | 0.1223 |
|                   | 3.157 | MAX | 0.1243 |
| 0.010<br>Oversize | 3.233 | MIN | 0.1273 |
|                   | 3.284 | MAX | 0.1293 |
| 0.020<br>Oversize | 3.360 | MIN | 0.1323 |
|                   | 3.411 | MAX | 0.1343 |
| 0.030<br>Oversize | 3.487 | MIN | 0.1373 |
|                   | 3.538 | MAX | 0.1393 |
| 0.040<br>Oversize | 3.614 | MIN | 0.1423 |
|                   | 3.665 | MAX | 0.1443 |

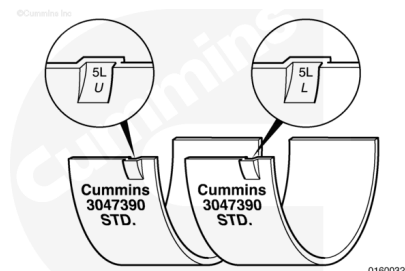


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## Install

**Note :** If bearings are to be reused, make sure they are put back in their original locations.

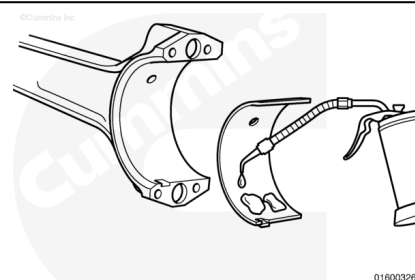
Part numbers of upper and lower new bearings **must** be identical.



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Lubricate the upper bearing with engine oil and install it in the connecting rod with the tang seated firmly in the machined slot.

**Note :** Rod bearings are available in oversizes of 0.010, 0.020, 0.030, and 0.040 inch. If rod journals have been ground undersize, the amount will be stamped on the counterweight closest to the front of the engine.

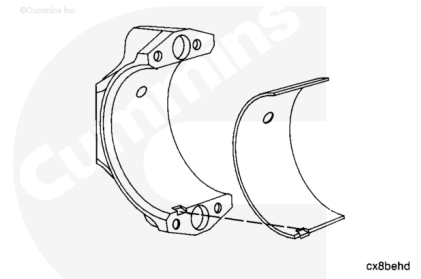


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**⚠ CAUTION ⚠**

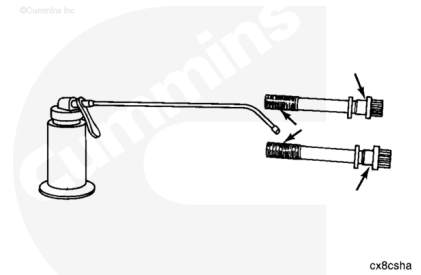
**Do not lubricate the back of the bearing shells.**

Install the lower bearing shell in the rod cap. Make sure the bearing tang is in place in the machined slot. Also, make sure the bearing ends are even with the parting surfaces.



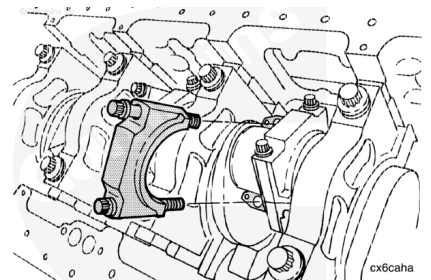
Lubricate the connecting rod capscrew threads and washers with engine oil and allow the excess to run off.

Install the washers and capscrews in the caps.



**⚠ CAUTION ⚠**

**Connecting rods and caps are not interchangeable, and caps fit in only one direction. Rods and caps are machined as an assembly and failure will result if they are mixed.**

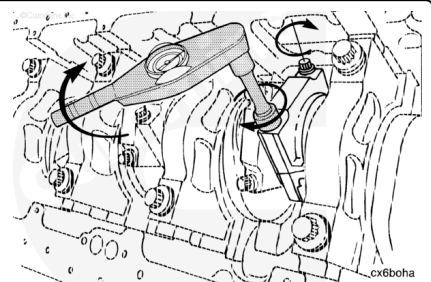


Install the connecting rod caps. Make sure the side of the cap with the chamfer is toward the fillet of the journal.

Use the following steps to tighten the capscrews. Apply torque plus angle to both capscrews on the connecting rod.

**Torque Value:**

1. 102 n•m [ 75 ft-lb ]
2. Rotate 90 degrees
3. Loosen both capscrews.
4. 102 n•m [ 75 ft-lb ]
5. Rotate 90 degrees.



**Note :** If it is not possible to use an angle gauge when applying the required torque to the connecting rod capscrews, the following sequence can be used:

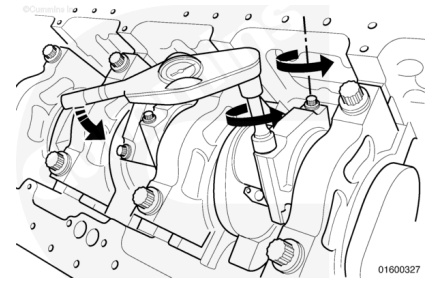
Use the following steps to tighten the capscrews.

**Torque Value:**

1. 108 n•m [ 80 ft-lb ]
2. 217 n•m [ 160 ft-lb ]
3. 325 n•m [ 240 ft-lb ]
4. Loosen both capscrews.

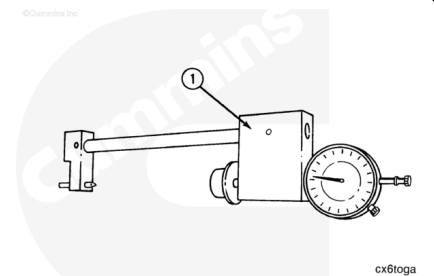
- 5. 108 n•m [ 80 ft-lb ]
- 6. 217 n•m [ 160 ft-lb ]
- 7. 325 n•m [ 240 ft-lb ]

Loosen both of the connecting rod capscrews.



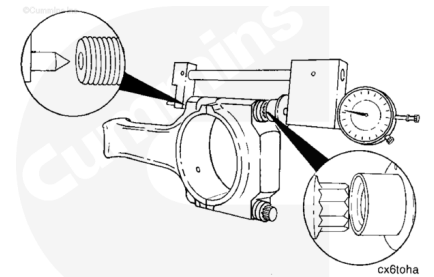
Use the connecting rod capscrew tool (1), Part Number 3822764, and the following procedure to measure the stretch of the capscrews.

**Note :** The use of two tools reduces the possibility of error and requires less time. The following steps show the usage of one tool.



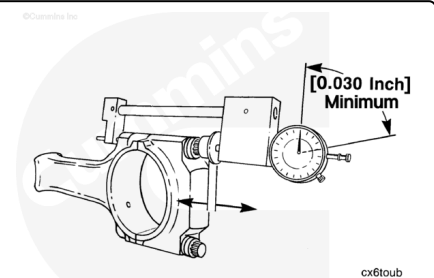
Position the tool on the first rod capscrew as shown.

Make sure the anvil is centered in the hole of the rod capscrew and the tool contacts the head correctly.



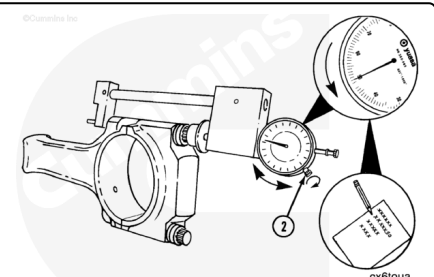
Check to make sure there is a minimum of 0.030 inch upward travel remaining on the indicator.

If there is **not** sufficient travel, loosen the set screw on the bottom of the tool and move the anvil downward.



Make sure the tool is in position and adjust the indicator dial until the needle points to the 0.050 inch mark.

Record the value as "first bolt zero". Tighten the lock (2) on the indicator dial to maintain the reading.

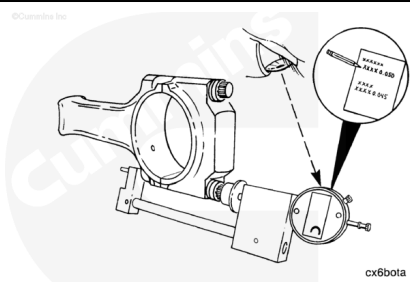


Remove the tool from the first reading and position it on the other capscrew of the same rod.

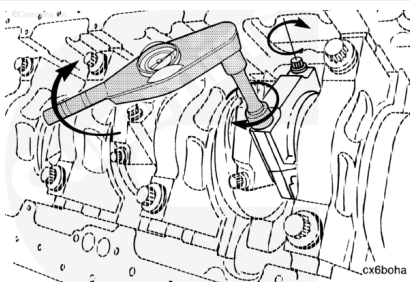
**Note :** There is no requirement for the capscrews to be the same length.

Record the reading obtained as “second bolt zero”.

Remove the tool.

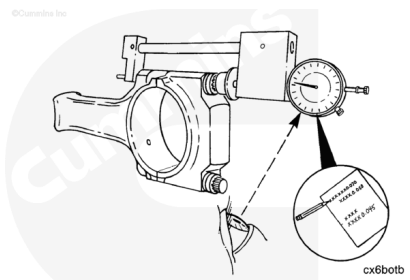


Tighten the capscrews alternately and evenly. Use the previous torque values in this procedure.



**Note :** Do not adjust the indicator dial.

Place the tool on the first connecting rod capscrew and take the reading 1.78 mm [0.070 in].

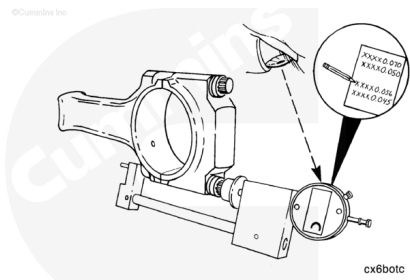


**Note :** Do not adjust the indicator dial.

Place the tool on the second rod capscrew and take the reading.

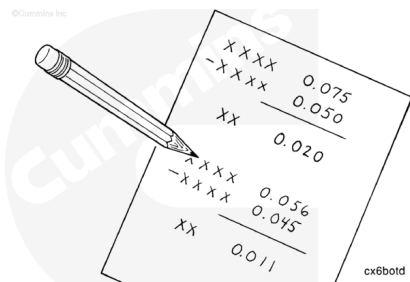
Record the value as “second bolt tight”. In this example, the value is 1.42 mm [0.056 in].

Remove the tool.



Determine the amount of capscrew stretch by subtracting the “bolt zero” from the “bolt tight” value for each capscrew.

Computing Capscrew Stretch



| Computing Capscrew Stretch |            |
|----------------------------|------------|
|                            |            |
| “First bolt tight” value   | 0.070 inch |
| “First bolt zero” value    | 0.050 inch |
|                            |            |



| Computing Capscrew Stretch |              |
|----------------------------|--------------|
|                            |              |
| "First bolt stretch"       | = 0.020 inch |
|                            |              |
| "Second bolt tight" value  | 0.056 inch   |
| "Second bolt zero" value   | 0.045 inch   |
|                            |              |
| "Second bolt stretch"      | = 0.011 inch |

Install the new capscrew using the specified torque steps and repeat the capscrew stretch measurement procedure, assuming the other capscrew was within specifications.

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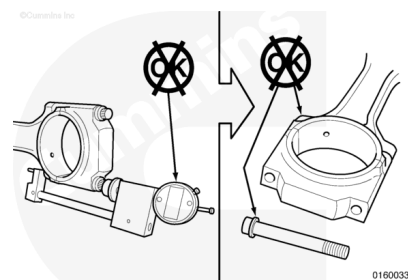
(-) .343 mm  
[.0185 in.]

365 Nm  
[270 FT. lbs.]  
Max

0160003

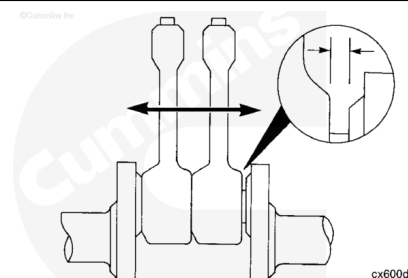
**Note :** If the capscrew that was tightened further to the maximum torque amount remains less than the

minimum stretch specification, the connecting rod and capscrew must be replaced.



## Side Clearance Check

- To measure the amount of connecting rod side clearance, move the rods as far as they will go toward the crankshaft web. Measure side clearance with a feeler gauge.
- Refer to Procedure 001-054 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)

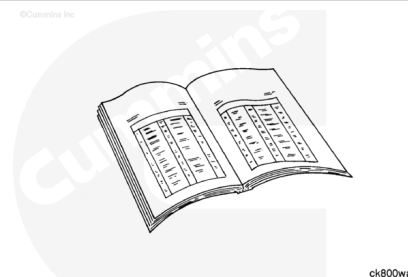


**Note :** Both rods must move freely from side-to-side.

## Finishing Steps

### **⚠ WARNING ⚠**

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



### **⚠ WARNING ⚠**

This component weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the oil suction tube and brackets. Refer to Procedure 007-040 in Section 7.  
(/qs3/pubsys2/xml/en/procedures/28/28-007-040-tr.html)
- Install the lubricating oil pan adapter assembly. Refer to Procedure 007-017 in Section 7.

(/qs3/pubsys2/xml/en/procedures/28/28-007-017.html)

- Install new full flow filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
- Fill engine with clean 15W-40 lubricating oil. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.